Figure 6. *Mytella strigata* biofouling green mussel farms in Bacoor City, Cavite, Manila Bay Photo from https://businessmirror.com.ph/2020/02/17/fake-tahong-invades-bacoor-mussel-farms/

5. Natural dispersal

Dispersal by purely natural means is not included as a pathway of biological invasions (Gaston 1996). Examples include range expansion by flight or any other medium of natural locomotion or transport. However if human created or crafted material is involved in rafting dispersal of IAS, then this may be considered as a case of biological invasion. The 2011 Great East Japan earthquake generated a large tsunami that caused an unprecedented biological transoceanic rafting event from the northwestern Pacific coastline of Japan towards North America on the eastern Pacific(Carlton et al. 2017). Millions of human made objects from small plastics to large docks and whole ships were cast adrift in the Pacific (Murray et al. 2018). This provided a substrate for biofoulers. Large debris could carry up to 20 to 30 mega-species of biofoulers (Carlton et al. 2017). These biofouled debris can constitute an IAS risk (Therriault 2017).

While a tsunami is a relatively rare event, a more common one is fouler dispersal by rafting on coastal currents of floating plastic debris, wood and, bamboo. Marine litter often originate from